

The Evolution of NYC Open Data

Ching-Yuan Peng, Jinu Hyun, Natiq Khan

Center for Data Science, New York University

{cp4516, jh10305, nak9135}@nyu.edu

Abstract

This paper replicates and modernizes the landmark 2014 landscape survey of NYC Open Data by Barbosa et al., revisiting the same portal a decade later to measure how the ecosystem has evolved in scale, quality, and integration potential. Using the Socrata SODA API, we collected metadata and sample values from 2,391 datasets and systematically compared our findings against the 2014 baseline. Our results show that while total dataset volume remained stable, the category composition shifted substantially — City Government and Education now dominate, displacing Social Services from the top position. Data quality improved: the proportion of tabular datasets increased from 75% to 88%, and the share of never-modified datasets declined from 70% to 60%. However, schema heterogeneity increased markedly, with the independence rate rising from 46% to 91% at Jaccard threshold 1.0, indicating that name-level matching has become less effective over time. To address this, we evaluated three complementary joinability detection methods — word-level Jaccard similarity, the Lazo MinHash-based algorithm, and vector embedding cosine similarity — against a stratified ground truth of 300 human-labeled column pairs. Lazo achieved the highest overall performance (F1: 0.90, Precision: 0.92, Recall: 0.90), outperforming both Jaccard (F1: 0.68) and embeddings (F1: 0.67), which converged to nearly identical results, suggesting that value-domain overlap is a more discriminative signal than column-name similarity for this dataset. We further synthesized the joinability relationships into a knowledge graph of 2,150 nodes and 182,098 edges, identifying key data hub datasets within the NYC ecosystem.

Keywords: open urban data, data integration, schema matching, joinability detection, MinHash, vector embeddings, Jaccard similarity, knowledge graph, NYC Open Data, longitudinal study

Introduction

The rise of open government data portals has created new opportunities for civic technology and data-driven policy. In 2014, Barbosa et al. (2014) conducted a landmark landscape survey of open urban data portals across nine major cities, establishing the first systematic empirical baseline for the field. Among all cities examined, New York City emerged as the most data-rich case, with 2,411 datasets and a notably high schema independence rate of 46%—meaning nearly half of all schemas could not be matched to any other dataset using exact column-name comparison.

More than a decade later, both the data and the available tools have changed substantially. The original study relied on Jaccard string-matching, a method blind to semantic equivalences like "boro" vs. "borough". This paper replicates and

modernizes that baseline using three complementary methods: Jaccard (reproduced as a direct baseline), vector embedding cosine similarity (semantic column-name matching), and the Lazo algorithm (MinHash-based column value overlap). Together, these span the lexical, semantic, and instance-level dimensions of dataset integration potential.

We focus on four research questions: (1) How has NYC Open Data’s scale and format composition changed since 2014? (2) Have null rates and schema quality improved? (3) Can semantic embeddings outperform lexical methods for joinability detection? (4) How do LLM-based, embedding-based, and MinHash-based methods compare for identifying joinable column pairs?

Related Works

The most direct antecedent to this work is Barbosa et al. (2014), who surveyed open urban data portals across nine cities using the Socrata platform. Their study established quantitative baselines for dataset volume, format composition, schema heterogeneity, and data quality—including the finding that 46% of NYC schemas were fully independent under Jaccard-based clustering. Our work directly replicates and extends their methodology for NYC, a decade later.

Rahm and Bernstein (2001) identifying semantically equivalent attributes across heterogeneous sources, which has a long history in database research. Early approaches relied on lexical similarity, structural cues, and constraint-based heuristics. Abedjan et al. (2015) surveys document the range of metrics used to characterize dataset structure and quality, many of which overlap with the 2014 paper’s quality indicators. These traditional methods serve as our lexical baseline.

Recent work applies pre-trained language models to table understanding tasks. Zhang et al. (2020) uses contextual embeddings for semantic column type detection; Suhara et al. (2022) extends this to multi-column relational annotation. Narayan et al. (2022) demonstrate that foundation models can perform schema alignment with minimal supervision. These works motivate our use of embedding cosine similarity as a semantic complement to Jaccard matching.

Identifying joinable tables across a data lake is an active research area. Nargesian et al. (2018) address table union search as a schema-level problem; Zhu et al. (2019) propose JOSIE, which uses set overlap to find joinable tables efficiently. Fernandez et al. (2019) takes an instance-level ap-

proach: it uses MinHash signatures to estimate Jaccard containment between column value domains, enabling detection of semantic joins even when column names differ entirely. We adopt Lazo as our third signal layer alongside lexical and semantic name-based methods.

Experiment Design

Dataset

Our dataset was collected from the NYC Open Data portal (data.cityofnewyork.us), the same Socrata-based platform used by Barbosa et al. (2014). We retrieved metadata and column-level information for all publicly available tabular datasets via the Socrata SODA API, yielding 2,391 datasets at the time of collection in early 2026. For each dataset, we collected: dataset-level metadata (title, description, category, view count, creation and update timestamps) and column-level metadata (column name, data type, and up to 100 non-null sample values per column).

It is important to note a fundamental limitation of our longitudinal comparison: the 2014 paper analyzed a snapshot of NYC Open Data as it existed in 2014, whereas our analysis reflects the portal’s current state in 2026. We do not have access to the original 2014 snapshot, and the portal does not maintain historical versioning of its catalog. Datasets from 2014 may have been deleted, renamed, or restructured; new datasets have been added. As such, our comparisons against 2014 baselines reflect changes in the overall ecosystem, not a controlled before-after study of identical datasets. This constraint is inherent to longitudinal open data research and is acknowledged as a boundary condition of our findings.

Method

Jaccard We apply Jaccard similarity at two levels, corresponding to two distinct analytical goals.

Schema-level Jaccard (replicating the 2014 baseline). Following Barbosa et al. Barbosa et al., 2014, we define each dataset’s *schema* as the set of its column names, lowercased. The Jaccard similarity between two schemas A and B is:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

Column-level Jaccard (joinability candidate generation). For the cross-dataset joinability task, we apply a word-level variant of Jaccard at the column-name level. Each column name is tokenized by splitting on underscores, spaces, and hyphens, with tokens of length ≤ 2 discarded. The similarity between two column names is then the Jaccard score over their respective token sets:

$$J_{\text{word}}(c_i, c_j) = \frac{|\text{tok}(c_i) \cap \text{tok}(c_j)|}{|\text{tok}(c_i) \cup \text{tok}(c_j)|}$$

Lazo To complement the name-based methods (Jaccard and Embedding), we use the Lazo algorithm to detect joinability at the instance level, identifying column pairs whose value domains meaningfully overlap, regardless of how the

columns are named. This catches joins that name-based methods miss entirely, such as a column named `boro` and one named `borough` which share identical value sets but have no lexical similarity.

For each column, we extract up to 100 non-null sample values and apply a cardinality filter, retaining only columns with 2–50 unique non-numeric values. Columns with a single unique value are constants and uninformative for joins; columns with more than 50 unique values are likely free-text or identifier fields whose sample-based overlap estimates are unreliable. Purely numeric columns are excluded because small integer sets (e.g. $\{1, 2, 3, 4, 5\}$) appear across unrelated columns, including age categories, district codes, and survey responses, and produce excessive false positives. After filtering, we retain roughly 38% of all columns as viable candidates for value-overlap matching.

Each viable column is encoded as a 128-permutation Min-Hash signature over its lowercased string values. We index all signatures using Locality Sensitive Hashing (LSH) to avoid brute-force $O(n^2)$ comparison across the full column space. For each candidate pair (c_i, c_j) returned by LSH, we compute asymmetric containment in both directions:

$$\text{containment}(c_i \rightarrow c_j) = \frac{|V_i \cap V_j|}{|V_i|} \quad (1)$$

and take the maximum of the two directions as the final score. The asymmetric formulation is intentional: a dataset covering only 3 NYC boroughs should still be flagged as joinable with one covering all 5, even though symmetric Jaccard would penalize the size difference.

We retain a pair as a candidate if its containment score is ≥ 0.5 and the columns share at least 3 distinct values in common. The minimum-overlap requirement guards against trivially small sets; for example, two unrelated yes/no survey columns would otherwise score 1.0 on containment despite having no semantic relationship. We chose 0.5 as the threshold to mean the same thing as the Jaccard threshold: the majority of one column’s value domain is contained in the other’s.

Embedding To capture semantic similarity between column names, we use sentence embeddings to complement the Jaccard and Lazo-based matching. We embed all 24,308 unique column names using sentence-transformers/all-MiniLM-L6-v2, which maps each column name to a 384-dimensional vector. Before computing pairwise similarity, we remove 3,677 non-interpretable or machine-generated columns using a regex filter. These include survey matrix codes, system-generated fields, numbered variants, and more. Such names often have similar alphanumeric structure but no semantic meaning, causing embedding models to assign misleadingly high similarity scores.

After pre-filtering, we compute the pairwise cosine similarity between every embedding, excluding self-comparisons:

$$S_{\text{emb}}(c_i, c_j) = \frac{\mathbf{v}_{c_i} \cdot \mathbf{v}_{c_j}}{\|\mathbf{v}_{c_i}\| \|\mathbf{v}_{c_j}\|}$$

We select embedding matches using a threshold of $S_{\text{emb}} \geq 0.85$. This threshold is higher than the Jaccard and Lazo thresholds because embeddings measure semantic proximity rather than exact name or value overlap. Consequently, they tend to assign high scores to columns that seem related but not actually joinable. We use a high threshold to reduce false positives.

The threshold was chosen by inspecting score bands. At 0.80, 31,349 column-name pairs were retained; at 0.85, this dropped to 19,465 pairs; and at 0.90, only 11,126 pairs remained. We use 0.85 as a compromise that preserves strong semantic variants while discarding the noisier pairs.

Ground Truth Labeling

Definition of Joinability A column pair (c_i, c_j) drawn from two different datasets is labeled as joinable if a domain expert would consider the two columns to refer to the same real-world attribute and their value domains to be meaningfully overlapping—such that a relational join on these columns would produce semantically valid results. Each pair was independently labeled by all three team members; the final ground truth label is determined by majority vote.

Candidate Generation Given the ~ 1.6 billion possible cross-dataset column pairs, exhaustive human labeling is infeasible. We first reduce the candidate space using the three methods described above: a pair is nominated if at least one method classifies it as a match under its respective threshold ($J_{\text{word}} \geq 0.5$, Lazo containment ≥ 0.5 , or embedding cosine similarity ≥ 0.85). The union of all nominated pairs constitutes the labeled pool.

Stratified Sampling Design To ensure that the ground truth captures both easy and hard cases—and supports unbiased evaluation of all three methods—we employ stratified sampling across six strata defined by method agreement:

Each stratum serves a distinct diagnostic purpose:

- **High** establishes a consistency baseline where all methods agree.
- **Medium (J+L)** tests whether Embedding misses value-level overlaps that lexical and instance-based methods catch.
- **Medium (J+E)** tests whether Lazo misses cases where column names are lexically and semantically aligned but values do not overlap.
- **Medium (L+E)** captures the rare case where value domains and semantics align but column names are lexically unrelated.
- **Low** probes each method’s unique detections—pairs flagged by only one signal.

- **Negative** is essential for computing Recall: without true negatives, false negatives cannot be estimated.

The Medium strata are the most analytically valuable: each Medium group isolates a specific method disagreement, allowing us to directly test whether a given method captures signal that the other two miss. The Negative stratum, sampled randomly from pairs rejected by all methods, is essential for computing Recall—without it, we cannot estimate how many true joinable pairs were missed entirely.

Results

Dataset Landscape and Quality

Figure 1 shows the 2026 category distribution of NYC Open Data with 2014 top-3 rankings annotated for reference. The composition has shifted substantially over the decade. City Government, ranked third in 2014, is now the largest category with 680 datasets; Education, absent from the 2014 top-three entirely, is the second-largest with 582 datasets. Conversely, Social Services and Housing & Development—the top two categories in 2014—have fallen to fifth and fourth place, respectively. This suggests that the portal’s growth has been driven primarily by administrative and civic transparency data rather than by the social-service datasets that dominated the earlier period.

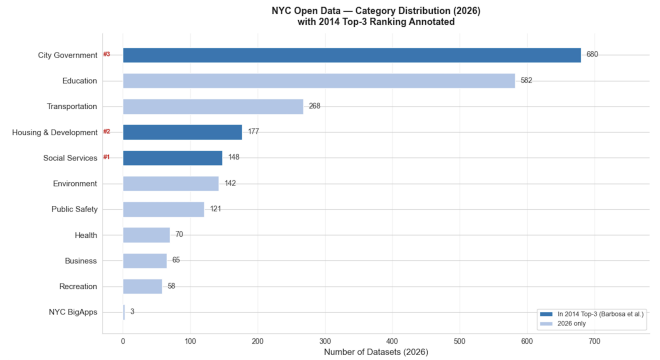


Figure 1: NYC Open Data category distribution in 2026, with 2014 top-3 rankings annotated. Exact 2014 counts were not published; only qualitative rankings are available from Barbosa et al.

Data quality also improved along several dimensions. The proportion of tabular datasets increased from 75% to 88%, reflecting either a deliberate curation effort or a shift toward structured administrative reporting. The share of datasets that have never been modified declined from 70% to 60%, suggesting more active maintenance. Null-value density likewise decreased, indicating that columns are more completely filled in the 2026 snapshot.

Schema heterogeneity, however, moved in the opposite direction. Figure 2 compares the schema independence rate—the proportion of unique schemas that have no exact column-name match in any other dataset—between 2014 and 2026. In 2014, Barbosa et al. reported that 46% of schemas were

fully independent; in 2026, that figure has risen to 91%, meaning that exact-match schema partners are now the exception rather than the rule. We attribute this primarily to the growth in average schema size (23.7 columns per dataset in 2026), which makes the probability of a complete column-name overlap combinatorially smaller, and to the influx of structurally novel categories such as Education and City Government that introduced schemas with no precedent in the 2014 catalog.

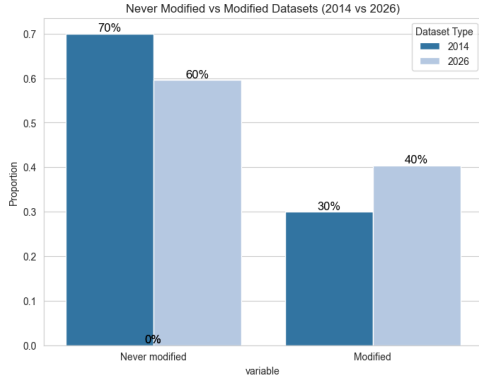


Figure 2: Schema independence rate in 2014 (Barbosa et al.) vs. 2026 (this study), at Jaccard threshold 1.0 (exact column-name match). Independence rose from 46% to 91%, indicating that name-level schema matching has become substantially less viable.

Model Comparison

We evaluate Jaccard, Lazo, and Embedding against the 300-pair stratified ground truth using Precision, Recall, and F1-score. Table 1 reports overall performance across all 300 labeled pairs.

Method	Precision	Recall	F1
Jaccard	0.68	0.68	0.68
Lazo	0.92	0.90	0.90
Embedding	0.67	0.67	0.67

Table 1: Overall Precision, Recall, and F1 across 300 labeled pairs.

Lazo substantially outperforms both name-based methods, achieving an F1 of 0.90 compared to 0.68 (Jaccard) and 0.67 (Embedding). Notably, Jaccard and Embedding produce nearly identical results across all metrics, suggesting that for this corpus, semantic embeddings capture the same signal as simple lexical token overlap.

Breaking performance down by confidence stratum reveals where these differences originate. In the High stratum (all three methods agree), all methods perform near-identically at $F1 \approx 0.98$, confirming that when methods converge, their predictions are reliable. Performance diverges in the Medium and Low strata.

In the Medium (J+E) stratum, where Jaccard and Embedding agree but Lazo does not, both agreeing methods achieve only $F1 = 0.48$. These are column pairs with similar names but non-overlapping value domains, representing the hardest cases to classify correctly.

In the Low stratum, where only one method flags a pair, Lazo maintains an F1 of 0.54 while Jaccard drops to 0.23. This confirms that value-domain overlap is a more robust standalone signal than column-name similarity when methods disagree.

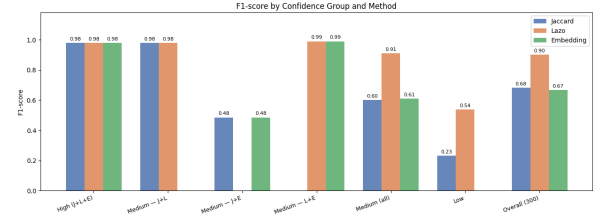


Figure 3: F1-score by confidence group and method.

Figure 3 shows F1-scores across all confidence groups. Figure 4 plots the overall Precision-Recall tradeoff. Lazo occupies the upper-right region of the PR space (high precision, high recall), while Jaccard and Embedding cluster together at lower performance, reinforcing the conclusion that they are functionally redundant on this dataset.

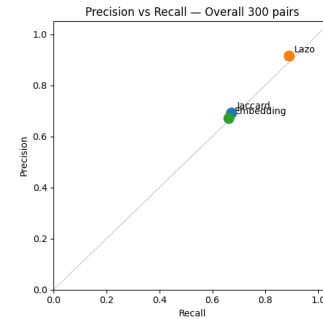


Figure 4: Precision vs Recall across all 300 labeled pairs.

Knowledge Graph

To summarize joinability at the dataset level, we construct a weighted knowledge graph from the merged column-pair output. Each row in this dataset is a candidate pair, with votes from the three joinability signals: Jaccard, Lazo, and Embedding.

We first retain only high-confidence column matches, defined as column pairs where at least two methods agree: vote count ≥ 2 .

This filtering step yields 267,526 matched column pairs. We then collapse these column-level matches into dataset-level relationships. If two datasets share one or more matched column pairs, we add an edge between them. The edge

weight is defined as the number of matched column pairs between the two datasets:

$$w(D_i, D_j) = \#\{\text{matched column pairs between } D_i \text{ and } D_j\}$$

This produces the full dataset knowledge graph with 2,150 connected dataset nodes, 182,098 dataset-to-dataset edges, and 15 connected components. The 2,150 nodes are the datasets that have at least one high-confidence relationship to another dataset.

Because the full graph is too dense to visualize, we construct a focused explanatory visualization. First, we rank datasets by weighted degree:

$$\text{weighted_degree}(D_i) = \sum_{D_j \in N(D_i)} w(D_i, D_j)$$

where $N(D_i)$ is the set of datasets connected to D_i . Degree counts how many datasets a node connects to, while weighted degree sums the strength of those connections. In this context, a dataset with high weighted degree is a strong hub: it is connected to many datasets through many matched columns.

For the focused visualization, we select four high weighted-degree hubs. For each hub, we select its 14 strongest neighboring datasets, where neighbor strength is measured by edge weight. Duplicate neighbors across hubs are merged. Finally, among the selected nodes, we draw up to 90 of the strongest edges. This gives a readable subgraph with 52 dataset nodes and 90 dataset relationships of the strongest dataset connections in the NYC Open Data ecosystem.

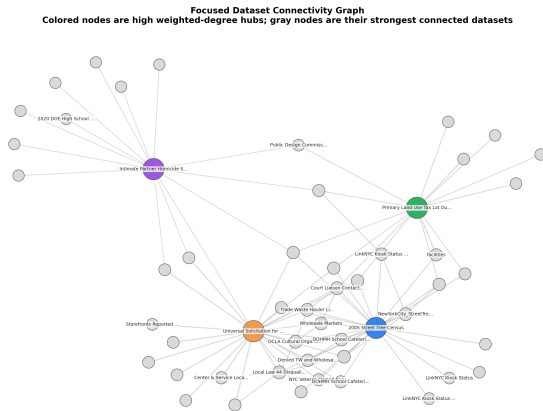


Figure 5: Focused dataset knowledge graph. Colored nodes are the selected hub datasets, gray nodes are their strongest neighboring datasets, and edges represent high-confidence dataset relationships derived from matched column pairs.

By tuning the parameters (# hubs, # neighboring datasets, # edges), one can add layers of complexity to the dataset network. Each visualization should be interpreted as an explanatory view rather than a complete map of all dataset relation-

ships. The full graph provides the quantitative network summary, while the focused graph highlights the main hub structure in a form that can be visually interpreted. The result suggests that a small number of datasets act as major connectivity hubs within NYC Open Data, linking many other datasets through shared or joinable schema elements.

Discussion

Schema Heterogeneity Increased, Not Decreased

Our finding that schema independence rose from 46% to 91% runs counter to the intuitive expectation that a maturing open data portal would become more standardized over time. We attribute this primarily to the growth in average schema size: at 23.7 columns per dataset in 2026, the probability of finding an exact full-schema match decreases combinatorially compared to the simpler schemas likely prevalent in 2014. Additionally, the influx of new dataset categories — particularly Education and City Government — introduced structurally distinct schemas that had no precedent in the 2014 catalog. This finding implies that schema-level exact matching becomes progressively less viable as open data ecosystems mature and diversify.

Value-Domain Overlap Is a More Discriminative Joinability Signal

Lazo’s substantial outperformance over both Jaccard and embedding-based methods (F1: 0.90 vs. 0.68 and 0.67) suggests that what columns are *called* is a weaker predictor of joinability than what values they *contain*. This is consistent with the known heterogeneity problem in municipal open data, where the same real-world attribute — borough, zip code, agency name — appears under dozens of different column labels across departments and datasets. Instance-level matching via MinHash signatures captures this structural overlap regardless of naming convention.

Semantic Embeddings Provided No Advantage Over Lexical Jaccard

The near-identical performance of embedding cosine similarity and word-level Jaccard (F1: 0.67 vs. 0.68) was the most surprising result of our study. One interpretation is that NYC Open Data column names are already sufficiently standardized — using common English words and consistent terminology — that there is little semantic gap for embeddings to bridge. Where Jaccard achieves high recall through token overlap, embeddings find the same pairs through semantic proximity, producing redundant rather than complementary signal. This has a practical implication: for well-curated municipal portals, the added computational cost of embedding-based matching may not justify its use over simpler lexical methods.

Implications for Open Data Integration Pipelines

Taken together, these findings suggest a two-stage architecture for practical dataset discovery: a fast lexical pre-filter

(Jaccard or embedding, which perform equivalently) to reduce the candidate space from billions to millions of pairs, followed by a Lazo-based value-overlap pass as the primary joinability signal. The knowledge graph derived from this pipeline — 2,150 nodes and 182,098 edges — demonstrates that meaningful integration potential exists across the NYC portal, concentrated around a small number of hub datasets with broad value-domain coverage.

Conclusion

This paper revisited the landmark 2014 landscape survey of NYC Open Data by Barbosa et al., replicating their methodology on a 2026 snapshot of the same portal and extending it with three complementary joinability detection methods.

Our longitudinal analysis reveals that the NYC Open Data ecosystem has grown more complex, not simpler, over the past decade. While the total number of datasets remained stable at around 2,400, the category composition shifted substantially toward City Government and Education content, the proportion of tabular datasets rose from 75% to 88%, and data quality improved as measured by null-value density. Most strikingly, schema independence increased from 46% to 91%, indicating that schema-level name matching has become progressively less viable as the portal has diversified.

For joinability detection, our evaluation on a stratified ground truth of 300 human-labeled column pairs demonstrates that the Lazo MinHash algorithm substantially outperforms both lexical and semantic name-based approaches (F1: 0.90 vs. 0.68 and 0.67), confirming that value-domain overlap is a more discriminative signal than column-name similarity in heterogeneous open data settings. The near-identical performance of word-level Jaccard and sentence embeddings further suggests that the added computational cost of embedding models may not be justified for well-curated municipal portals.

Finally, synthesizing high-confidence column matches into a knowledge graph of 2,150 nodes and 182,098 edges reveals that integration potential within NYC Open Data is real and concentrated: a small number of hub datasets link large portions of the portal through shared value domains.

Limitations and Future Work

Our study is constrained by the absence of the original 2014 data snapshot; all comparisons are against published statistics rather than a controlled before-after study of identical datasets. Future work should prioritize maintaining versioned portal snapshots to enable true longitudinal tracking. Additionally, our ground truth covers only 300 pairs sampled from method-nominated candidates, leaving unjudged pairs that may contain missed joinable relationships. Extending the ground truth with active learning or crowdsourcing, and applying the pipeline to other open data portals, would strengthen the generalizability of these findings.

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